Lecture Notes on

Semester #1 Review - Jeopardy Topics

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These notes summarize the key concepts reviewed during the Semester #1 Astrophysics Club Jeopardy meeting. They consolidate major ideas from our first four meetings.

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1 Scale of the Universe

1.1 Light-Year

- A light-year is a unit of **distance**.
- Defined as the distance light travels in one year.
- Since $c \approx 3.0 \times 10^8$ m/s,
$$1 \text{ light-year} \approx 9.46 \times 10^{15} \text{ m}.$$
- Used for interstellar distances.

1.2 Speed of Light

- Constant:
$$c = 299{,}792{,}458 \text{ m/s}.$$
- Maximum speed information can travel.
- Central to relativity.

1.3 Astronomical Unit (AU)

- Average Earth–Sun distance.
- 1 AU $\approx$ 93 million miles.
- Useful for planetary systems.

1.4 Parsec

- 1 parsec $\approx$ 3.26 light-years.
- Derived from parallax measurements.
- Common professional astronomy unit.

1.5 Andromeda Galaxy

- Closest major galaxy to the Milky Way.
- $\sim$ 2.5 million light-years away.
- Will merge with the Milky Way in $\sim$ 4 billion years.

2 Celestial Motion

2.1 Geocentric vs Heliocentric Models

- Ancient model: Sun orbits Earth.
- Copernicus: Earth orbits Sun.
- Kepler refined heliocentric model.

2.2 Kepler's Laws

- Planets orbit in ellipses.
- Equal areas swept in equal times.
- $T^2 \propto a^3$.

2.3 Newton's Gravity

$$F = \frac{Gm_1m_2}{r^2}.$$

- Gravity treated as a force.
- Explains Kepler's laws mathematically.

2.4 Einstein's View

- Gravity is curvature of spacetime.
- Mass bends spacetime.
- Objects follow curved geometry.

2.5 Dark Energy

- Observed accelerated expansion of universe.
- Acts like negative pressure.
- Dominates large-scale cosmic dynamics.

3 Stars

3.1 The Sun

- Closest star to Earth.
- Main sequence star.
- Powered by hydrogen fusion.

3.2 Constellations

- Patterns of stars.
- Appear connected from Earth's perspective.
- Not physically close in 3D space.

3.3 Nuclear Fusion

- Combines hydrogen into helium.
- Releases energy via $E = mc^2$.
- Occurs at $\sim 10^7$ K in stellar cores.

3.4 Nebulae

- Giant clouds of gas and dust.
- Sites of star formation.
- Can also be supernova remnants.

3.5 Supernovae

- Massive stellar explosion.
- Occurs when core collapses.
- Produces heavy elements.
- Can leave neutron star or black hole.

3.6 Neutron Stars

- Extremely dense stellar remnants.
- Supported by neutron degeneracy pressure.
- A teaspoon can weigh billions of tons.

3.7 Black Holes

- Form from collapse of massive stars.
- Event horizon marks point of no return.
- Characterized by Schwarzschild radius:

$$R_s = \frac{2GM}{c^2}.$$

4 Big Picture Connections

Across all meetings:

- Gravity governs structure.
- Fusion powers stars.
- Mass determines stellar fate.

- Expansion shapes cosmic evolution.
- Observations rely on light.

Astrophysics connects:

- Mechanics
- Thermodynamics
- Electromagnetism
- Quantum physics
- Relativity

Semester #1 built the foundation for deeper topics ahead.